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THE ECONOMIC IMPACTS OF UBER SERVICES ON TRADITIONAL TAXI OPERATORS IN THE ACCRA METROPOLIS

MANUEL NII MARTEY MENSAH

2018

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UNIVERSITY OF CAPE COAST

THE ECONOMIC IMPACTS OF UBER SERVICES ON TRADITIONAL TAXI OPERATORS IN THE ACCRA METROPOLIS

BY

Manuel Nii Martey Mensah

A Thesis submitted to the Department of Geography and Regional Planning, Faculty of Social Sciences and University of Cape Coast, in partial fulfilment of the requirement for the award of a Bachelor of Science Degree in Geography and Regional Planning.

JUNE, 2018

DECLARATION

I, Manuel Nii Martey Mensah, hereby declare that this dissertation is my own original work. To the best of my knowledge, this is not material previously published by another person for the award of any degree of the University or any other institution or for public consumption, except where due acknowledgement has been made.

Student signature……. Date……

Supervisor's signature……. Date……

Supervisor: Dr. (Mrs.) Regina Obilie Amoako-Sakyi

ABSTRACT

The introduction of a ride-hailing service which deals with driver-rider interaction through a mobile application has made the request for taxis much easier. Traditionally, one has to stand by the road to call for a taxi and negotiate prices. The introduction of Uber services has changed the face of transportation in Accra and Kumasi. The study which used an exploratory design collected data from 417 respondents who were sampled through e-snowballing and e-accidental sampling techniques. An online questionnaire was used to collect data from Uber riders while an electronic interview guide was employed for Uber and taxi drivers. Analysis of the sample data revealed that, Uber comes as a disruptive business and not a destructive one as the study shows that jobs of taxi drivers have not been taken from them but rather an alternative to taxis has been created which makes it a matter of choice for drivers and riders to join the technological trend or not. It also discovered that wages of taxi drivers had been negatively affected since the inception of the ride-hailing service. However, 65% of taxi drivers said they were not willing to adopt the pricing system and this was due to ignorance on the side of most taxi drivers on the operations of Uber. Uber's percentage on each ride makes Uber drivers resort to cheating the system. It is therefore recommended that Uber revises its percentage take per trip. Also, a regulatory body by the government needs to be established to regulate the fares of both businesses to ensure a healthy competition.

KEY WORDS

Capacity utilisation

Disruptive business

E-accidental sampling technique

E-snowballing technique

Ride-hailing

Taxi

Uber

ACKNOWLEDGEMENT

An immeasurable depth of gratitude goes to God for making this project a success. My most sincere gratitude goes to Dr. (Mrs.) Regina Amoako-Sakyi for having strong interest in the topic which translated to the guidance she gave and new things I kept learning from her. I will like to register a heart- felt gratitude to my parents, Mr. Raymond Mensah & Mrs. Grace N. D. Mensah, for financing this project and for their encouragement. A big appreciation to my social media friends whom without hesitancy contributed tremendously to the success of this project.

DEDICATION

This work is dedicated to Uber, their riders and drivers and to all taxi drivers in Accra.

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LIST OF ACRONYMS

GPRTU: Ghana Private Roads Transport Union

ICT: Information and Communications Technology

ILO: International Labour Organisation

SPSS: Statistical Package for Social Sciences

CHAPTER ONE

INTRODUCTION

1.1 Background to the study

The introduction of a private taxi service which deals with driver-rider interaction through a mobile application has made the request for taxis much easier. Traditionally, one has to stand by the road to call for a taxi and negotiate prices. The introduction of a transport service where prices are predetermined makes the advent of Uber services into the country a huge changing phase of the transportation industry in Accra and Kumasi.

Uber belongs to the concept of a shared economy in the transport industry. Belk (2014) describes the sharing economy as a phenomenon born out of the internet age. A sharing economy is an umbrella concept that encompasses several ICT developments and technologies which endorses sharing the consumption of goods and services through online platforms (Hamari, Sjöklint, & Ukkonen, 2016). A sharing economy is therefore, any cost-effective activity involving online transactions. The term Uberisation has been coined out of this phenomenon to mean an economic system where agents exchange under-utilized capacity of existing or human resources.

The Uber service makes use of a digital platform where patrons of the service known as riders request for taxis directly from drivers while being shown the distance, estimated driver's time of arrival, estimated ride time and fare. It runs on an optimization software where distance and time are used in the determination of fares for riders. From observation and self-experience with the service, it can be deduced that the rates, although dynamic with changes in demand and supply, are mostly

cheaper than that of the traditional taxis. For instance, if there is high demand by riders but few drivers on the road, the prices increase and people are left with no choice but to go with the traditional taxis.

Around the world, Uber has been a sensation in the development of transportation. Though it is progressively growing, it has an adverse impact on traditional taxi operators in countries where they are. Currently, Uber is facing several challenges in most European countries and in the United States where they operate. These are challenges they face from traditional taxi operators mostly. Reports of a strike action by taxi drivers in the United States in January 2017 which encouraged users to stop using the Uber app with an online campaign #DeleteUber (Delete Uber). Another was a violent strike action in France by taxi drivers, smashing cars, setting tires on fire and blocking traffic to register their displeasure at the operation of Uber in France, the Chinese government banned drivers of private cars from offering their services through taxi-hailing apps. According to Seattle Times, the Uber offices in Guangzhou were raided and several equipment used in running the service were seized. These are few examples of incidents recorded in some countries around the globe especially of calls to boycott the Uber service (The Economist, 2017; The Seattle Times, 2015).

Uber's image was further dented when in March 2015 UN Women backed out of an Uber partnership which was to create jobs for a million women drivers by 2020, after trade unions and civil society groups and some governments opposed to the scheme.

Uber interestingly owns none of its vehicles used in running the service, but signs up private drivers willing to provide rides to paying passengers and passes the ride requests directly to them (Petropoulos, 2016). The vehicles used belong to Uber

drivers, individuals or car renting and leasing companies who give their cars to others for use in which they earn some income from. It can therefore be hypothesized that the advent of Uber has had a significant impact on traditional taxi operators, raising competition in the industry and has provided an alternative livelihood for drivers and car owners who lease their cars out for the operations as many of them take.

Uber was introduced in Accra, Ghana on 8th June 2016 and began its full operations on 11th June, 2016, making it the 8th African country to be introduced to the service. It set its base fare at GH¢1.70 + GH¢1.30 per kilometer and GH¢0.19 per minute, making its minimum fare GH¢5 with a cancellation fee of GH¢5 which is added to the actual cost of a riders next patronage. It expanded its service to Kumasi on August 31, 2017.

This study borrows from the economic variables used by Chang (2017) which he discusses in his paper, *The Economic Effects of Uber on Taxi Drivers in Taiwan,* to look at the impact on traditional taxi operators in the Accra Metropolis.

1.2 Problem Statement

Uber services around the world face a common problem in most countries in which they operate - the protest and agitations of traditional taxi operators against an unhealthy competition. In a personal interaction with 10 taxi drivers in the Accra Metropolis, selected at random, 7 of them complained that Uber operations in the city had snatched their customers from them with the other 3 not knowing what Uber is. While Uber is creating jobs, it is also having an effect on an already existing industry. This study therefore sought to identify the major effects Uber has had on traditional taxi operators in the Accra Metropolis.

1.3 Research Aim and Objectives

The aim of this study was to examine the economic impact of the Uber service on traditional taxi operators in the Accra Metropolis.

The specific objectives of this study was to:

i. Establish the Uber impact on taxi drivers' market power by measuring the profit margins of traditional taxi operators.

ii. Measure the impact of Uber on traditional taxi operators with different operation models – the unionized taxi operators and the non-unionized taxi operators.

iii. Enquire if traditional taxi operators are willing to adapt the Uber system of pricing.

iv. Investigate the factors which influence people's patronage of the service.

v. Examine whether the introduction of Uber has had an effect on the operating costs of traditional taxi operators.

vi. Determine the categories of people who patronize the service by taking into consideration their age, occupation and sex.

1.4 Justification of the Study

Uber is creating a growing impression in the country hence the need for this study. Most of the studies regarding Uber's economic impact on the traditional taxi industry have been done in some parts of the United States, Europe, Asia and Africa where Uber operates. This study therefore is expedient since traditional taxi operators in Accra have begun complaining of an unfair competition by Uber just like it is with their colleagues around the world. This study will therefore aid the various stakeholders in the transport sector to regulate and harmonize the operations of Uber and traditional taxis in order to avoid future incidences of riot or strike actions as they occur in some parts of the world where Uber operates.

1.5 Limitations of the Study

i. A majority of taxi drivers especially the stationed ones refused to partake in the process by the mere mention of Uber.

ii. Financial constraints on the side of the researcher rendered him incapable of covering larger numbers Uber and taxi drivers. Furthermore, some interviews had to be abrogated abruptly due to the relatively short distance the drivers and researcher had to cover.

iii. Some questionnaires were partially filled as stationed taxi drivers who offered to partake in the study had to leave once their vehicles were fully boarded by passengers. Some Uber drivers also had pending ride requests so had to move the moment we got to their destinations.

iv. Some drivers – Uber and taxi drivers, displayed coldness while being interviewed which in a way affected the quality of the interview since the questions had to be rushed through.

1.6 Outline of Project Report

The next chapter will discuss available literature relevant to this study. The subsequent chapter describes the methodology which talks about the data collection procedures employed. Data analysis will be presented in the following chapter. Eventually recommendations and concluding remarks and for the purpose of this study will be presented in the last chapter.

CHAPTER TWO

LITERATURE REVIEW

2.0 Introduction

The economic impact of Uber services on the traditional taxi industry is a growing concern not just in the Accra Metropolis but around the world. This literature review will therefore be looking out for what the economic impacts of Uber on traditional taxi operators constitute.

A number of researches have been carried out in parts of Europe, Asia, America and Africa based on this subject of ride sharing or ride-hailing by looking at different angles of the effect of Uber on taxi operators. Ride-hailing is the process of arranging a ride mostly privately owned cars through a mobile application. Ride-hailing is defined as, "of or relating to a car service with which a person uses a smartphone app to arrange a ride in a usually privately-owned vehicle." (*Dictionary.Com., 2018*). The term originates from the concept of stopping vehicles by lifting a hand and signalling that you want to be picked known as hailing a taxi. However, this has been borrowed in the Uber concept, where ride-hailing now refers to one being able to request for a car in the comfort of one's room and the car arriving at the destination of request.

This transportation revolution has had an effect on the taxi industry. In their paper, "Disruptive Change in the Taxi Business: The Case of Uber", Kreuger & Cramer (2015) stated that "The innovation of ride sharing services, such as Uber and Lyft, which use internet-based mobile technology to match passengers and drivers, is providing unprecedented competition in the taxi industry" (2015:2). They attributed

the high utilisation of UberX, an extension of Uber's service, to four major factors - Uber's more efficient driver-passenger matching technology; Uber's larger scale, which supports faster matches; inefficient taxi regulations; and Uber's flexible labour supply model and surge pricing, which more closely match supply with demand throughout the day. In their study, five cities were considered: Boston, Seattle, Los Angeles, San Francisco and New York City, for which a random sample of 2,000 drivers was selected for each city. Their findings depict UberX drivers have a higher capacity utilization rate than taxi drivers.

2.1 Employment Effect of Uber on Traditional Taxi Operators

Berger, Chen, & Frey (2017) classify the economic impacts of Uber on taxi operators into two – the employment impact and the wage impacts. According to their study in three cities - Los Angeles, San Francisco and New York, the evolution of taxi employment has taken a nose dive. For instance, the study depicts that the labour supply of taxi drivers saw a downward decline, prior to Uber's introduction in 2012, in Los Angeles but saw a sharp upward break after its introduction, with particular growth among the wage employed that mirrored similar but less obvious patterns in San Francisco. In New York, however, the labour supply taxi drivers decreased considerably after the introduction of Uber in 2011 though it recovered over subsequent years. According to the literature, their findings do not show that the introduction of Uber has had any effect on the employment of taxi drivers.

2.2 Wage impact of Uber on traditional taxi operators

The wage impact according to the work done by Berger, Chen & Frey (2017) show that although taxi drivers have had their jobs untouched, the income had been greatly affected. They stated that, "the hourly earnings of self-employed taxi drivers increased after Uber's introduction in contrast to the earnings of wage-employed taxi drivers" (2017:8). Hence, while waged drivers saw a decline in their income levels,

self-employed drivers saw a massive increase in their income. Machado & De Oliveira (2017) believe the impact of Uber on traditional taxi operators are minimal or non-existant. They stated in their literature, "The resistance by taxi drivers does not make much sense in theory and was not expected a priori, as they can arbitrate and choose between working in a taxi with or without an application or even more than one application" (2017:17). They however agreed that Uber's pricing quotation and transparency in regulations affects taxi drivers.

A study by Wallsten (2015) showed that taxi drivers in New York and Chicago responded to the competition by improving quality. He showed in its findings that spoilt credit card machines, rudeness, and talking on cell phones - all seem to have decreased along with Uber's rise. Hung-Hao (2017)'s study also affirmed that taxi drivers are finding ways of increasing income by means of having advertisements in or on the taxi cabs.

2.3 Pull factors of Uber patronage

The increase in the patronage of Uber services, hence, the rise in the income of self-employed taxi drivers is attributed to five factors - tariffs, practicality, trust, comfort, and safety. According to Natyari & Pradana (2016), these five factors were identified as the influencers of consumers' choice of Uber in the city of Bandung with comfort being the dominant factor. Consumers liked the transparency of the details and how fair the charges of the service were. Practicality refers to the ease with which customers access a service. With one being able to request for a ride through a mobile app in the comfort of his or her room or irrespective of the location, consumers prefer to ride with Uber. Consumers' trust in the service - ability, integrity and attitude of the company when dealing with consumers, which implies that customers feel safe and willing to rely on the company. The comfort of passengers is considered the most

dominant factor as to why Uber is patronized the most in Bandung and this is linked to the safety of customers.

This literature review has thrown some light on Uber's impact on taxi drivers. However, existing literature fails to mention if unionized and non-unionized taxi drivers suffer the same fate and it is also silent on the economic class of people who patronize the services of Uber. Hence, we cannot deduce if the income level of a person is an influence to Uber patronage.

This study investigated if these findings - the impact of Uber on taxi operators as found in existing literature is same for the Uber-Taxi competition within the Accra Metropolis.

CHAPTER THREE

RESEARCH METHODOLOGY

3.0 Introduction

This chapter describes the methods used in collecting data for the study, the instruments used, the population and sample targets, the study area and what type of study was employed.

3.1 Research Design

The research adopted an exploratory approach as it sought to find answers to the level of impact Uber has had on traditional taxis; why people prefer Uber to taxis; the demographics of Uber patrons; and which groups of taxi operators are affected the most by the operations of Uber services. Exploratory research is defined as a type of research usually conducted to study a problem that has not been clearly defined yet (Dudovskiy, 2018). The advantage of using this design is its adaptability to change and most importantly it sets basis for future studies.

3.2 Study Area

The Accra Metropolis is located in the Greater Accra Region, Ghana with an area of 894 km^2 and an elevation of 61m above sea-level. It serves as the economic, administrative and political capital of Ghana. Accra has an urban population of 2.27 million out of the about 4 million inhabitants in Greater Accra Region (World Population Review, 2018). The public transportation system in Accra include taxis, intra-city buses – the private ones known as *trotros* and the government operated Bus Rapid Transit (popularly called *Aayalolo*) which currently operates only one route and the intra-city Metro Mass Transits bus services known as *Kuffuor buses*, and the rail transport. The most used however are the intracity buses and taxis. Uber comes as a

recent addition to the public transport system and Accra is amongst the two cities currently operating the Uber service in Ghana, with the other being Kumasi. The study chose Accra as its study area considering the fact that Uber has been operating in Accra for a longer period as compared to Kumasi.

3.3 Population

The population of this study included people who patronise taxi and Uber's services irrespective of their age, religion, sex, employment status or educational level. It also targeted drivers of Uber and taxis in Accra who are of legal driving age irrespective of their sex and educational level.

3.4 Sample Size and Sampling Procedure

Uber has a population of 130,000 active riders and 3,000 active drivers in Ghana. Using a sample size calculator with a confidence level of 95% and error of margin of 5%, this study used a sample of 354 Uber riders and 43 Uber drivers. The challenge with this sample size is that, the population of Uber riders and drivers as stated above covers both areas of Uber operations (Accra and Kumasi). However, this study covered only Accra.

Among the traditional taxi drivers, the study focused on both unionised and non-unionised drivers in Accra and therefore a total of 20 taxi drivers were drawn at various stations including the Ministries Taxi Rank, Dansoman Roundabout Taxi rank and Kaneshie Stations and as well as non-stationed or "roaming" drivers.

The sampling technique used for Uber riders was that of an electronic snowballing technique. Here, the link to the online survey was shared via social media and respondents were encouraged to share amongst their contact list most especially people who had used Uber before and in Accra.

The disadvantage of using this technique is the tendency of people partaking in the survey more than once. Also, people who may not qualify to be part – that is, those who might use Uber but outside the study area and those who may be inquisitive about the study but have not patronised Uber before, may end up being part of the respondents.

The advantage of using this technique is that, it is able to cover a larger area despite Uber riders being scattered in the study area. The link to the online survey was activated for a period of two weeks and was closed after 91.7% of the calculated sample size was gathered and there was inactivity on the survey page after 5 days.

The accidental sampling technique was used for sampling taxi drivers. This technique had the researcher stopping taxi drivers at random. The technique was employed especially for non-stationed taxis. The disadvantage of this technique is the cost involved with regards to taxi fares – the researcher had to join the taxi to enable him administer the questionnaire.

The stationed taxis had the simple random sampling being applied on them. The researcher interacted with taxi drivers at random at some selected taxi ranks in Accra.

Uber drivers were sampled by use of a technique which the researcher defines as an Electronic Accidental Sampling (e-Accidental Sampling). This type of sampling had the researcher requesting for rides via the Uber app. Drivers who picked up the researcher partook in the study. This type of sampling saved the researcher from the inconvenience of scouting for Uber drivers since Uber drivers have nothing to identify them or their cars as Uber resources. The disadvantage however with this e-sampling technique is the probability of getting the same driver twice or more. In order to avoid this, the researcher made sure the respondent (driver) is a distance away from the

researcher's location - a radius of 10 metres away from the researcher was a best option. Just like the accidental sampling employed for taxi drivers, this is also cost involving with regards to transport fares and internet bundles in order to utilise the Uber app.

3.5 Research Instrument

The research instruments used were online surveys and electronic interview guides. The online survey was used for Uber riders since it is assumed patrons of this service are technologically inclined. Google Forms was the online platform used in designing and collecting the data. The instrument (Appendix 1) was disseminated via major social media platforms such as Facebook, Twitter, WhatsApp and Instagram.

The online surveys consist of two sections – Demographics and Uber Usage. The former collected details of the respondent and the latter collected data on the respondent's usage of the Uber service. The challenges with this instrument however is its inability to get geographic locations of respondents to verify if indeed they are in Accra. It also restricted the researcher from making further enquiries and clarification to responses derived. The major disadvantage of using this tool was the data was skewed towards the youth and tended to restrict respondents who are not tech-savvy, notably among these was the older generation. The advantage of using this instrument is it saved cost and since Uber users and riders are spread across, it was easier to reach them via social media.

A data collection mobile application, Kobocollect, was solely used to collect data from Uber and taxi drivers in Accra. Kobocollect is an open source tool used by humanitarians, development practitioners, global health workers, and researchers around the world (Kobo Toolbox, 2018). It is an application that enables a researcher to carry out a paperless survey. Kobo Toolbox allows the researcher to run quick

analysis on their platform and it is much effective if a laptop is used. The KoboCollect app allows the researcher to create multimedia surveys for data as well as geo-tag locations via the Global Position System which allows the researcher to known where and when the data was collected (Meier, P. 2012).

The electronic interview guide (Appendices 2 and 3) on KoboCollect consisted of two sections - the demographics and usage of their service. This was used because it was assumed by the researcher that the level of participation in an online survey by taxi and Uber drivers might be low due to their busy schedules and to some extent, their literacy level. Hence, it demanded the researcher adopting a face-to-face interaction with them.

The advantage of using this interview guide was it enabled the researcher to ask further questions on points raised and also helped seek clarification on some answers given and respondents gave information very essential to the topic. It was also economical for the researcher in terms of the reduction of cost involved in printing out questionnaires. It was convenient for the researcher since he did not need to carry a lot of papers.

The disadvantage of using this instrument is the draining of phone's battery by the application over a period of time. This challenge was however circumvented with the use of spare batteries and power banks to extend the survey duration.

3.6 Data Collection

The data was collected in two forms – online survey and questionnaires. The online survey was disseminated using social media platforms through a somewhat snowballing technique. The snowballing technique was as a result of respondents sharing the link to the online survey to people, mostly colleagues, whom they knew or assumed patronised the service. This helped garner a lot of responses from the

target respondents – Uber riders. The intention of the study was stated with the details of the researcher and the supervisor and those who consented to taking part of the survey went ahead to fill the form. The online survey link was opened for a period of two weeks and closed when enough responses had been accumulated.

Data was collected from taxi and Uber drivers through an electronic interview guide, as stated above, for a duration of two weeks on three days within the week – Fridays, Saturdays and Sundays, by boarding their vehicles and interacting with them. These days were chosen because they were convenient for the researcher per the researcher's schedule within the week. The researcher sought their consent to be able to carry out the interview.

KoboForm, which is the online questionnaire created, was downloaded via the app for offline use which is referred to as KoboCollect. After collecting the data needed, the form was saved and uploaded on the Kobo Toolbox online platform where the data, like Google Form, was tabulated onto a Microsoft Excel spreadsheet which the researcher can download to run his own analysis. The study employed focus group discussions and interviews, to collect data from taxi driver.

3.7 Data Processing and Analysis

The Statistical Package for Social Sciences (SPSS) version 16 was used for the input and analysis of the data collected from the field and employed the use of charts and cross-tabulations mostly in analysing the data.

CHAPTER 4

RESULTS AND DISCUSSIONS

4.0 Introduction

This chapter discusses the results of the analysed data. It therefore discusses the economic impact of Uber on traditional taxi operators in the Accra Metropolis.

4.1 Socio-demographics

The socio-demographic characteristics of the study participants on age, sex, educational level, occupations and number of dependents of the respondents, are presented in Tables 1 to 6.

4.1.1 Age

Table 1: Age distribution of respondents

	Percentages		
Age	**Uber Riders**	**Uber Drivers**	**Taxi Drivers**
10 - 20	4.5	2.5	0
21 - 30	78.8	55	20
31 - 40	15.5	40	20
41 - 50	0.9	2.5	45
51 - 60	0.3	0	15
Total	**100**	**100**	**100**

Source: Fieldwork (2018)

The data from respondents (Table 1) indicate Uber riders between 21 and 30 years patronize Uber's services the most (78.8%), making Uber's customer base a youthful

one. This could however be as a result of the research instrument and data collection employed which makes the data skewed towards the youth who are more tech-savvy.

People between the ages of 21 – 30 (55%) and 31 – 40 (40%) were also found to be the most employed by Uber as the stated year groups show a high representation of Uber drivers within these ages. This implies, within our respondents, Uber has more of both youthful riders and drivers. However, as at the time of conducting this survey, Uber had rolled out a new insurance policy for its drivers which barred drivers below the age of 25 from driving.

Most taxi drivers were however found to be between 41 – 50 years (45%) and above 50 years (15%). This means we have more of the older generation driving taxis.

4.1.2 Sex

Table 2: Sex distribution of Uber riders

	Percentages		
Sex	**Uber Riders**	**Uber Drivers**	**Taxi Drivers**
Male	52.5	100	100
Female	47.5	0	0
Total	**100**	**100**	**100**

Source: Fieldwork (2018).

Table 2 shows the sex distribution of Uber riders. Out of 354 respondents for Uber riders, the study had more male riders (52.5%) than female riders (47.5%).

For Uber drivers, all the 42 respondents were males (100%). This is not to say there are no female Uber drivers but the sampling technique employed did not get the researcher coming across female Uber drivers. The same applies to taxi drivers, as the study had an all-male response. This study therefore assumes there are more men involved in commercial driving than women in Accra Metropolis.

4.1.3 Occupation of Uber riders

Table 3: Occupation of Uber riders

Occupation	Frequency	Percent
Administration Professional	10	2.8
Business Professional	15	4.2
Business Service Agents	7	2.0
Cultural Professional	14	4.0
Engineering Professional	16	4.5
Finance Professional	37	10.5
Health Professional	15	4.2
ICT Professional	8	2.3
Marketing Professional	11	3.1
National Service Personnel	42	11.9
Other	47	13.3
Social Professional	6	1.7
Student	108	30.5
Teaching Professional	12	3.4
Unemployed	6	1.7
Total	**354**	**100.0**

Source: Fieldwork (2018).

One of the objectives of the study was to know the occupation of Uber patrons. This variable was used as a proxy for the socio-economic status (SES) of respondents. With the help of the International Standard Classification of Occupation (ISCO-08) by the International Labour Organisation (ILO, 2012), the occupations of respondents were categorised into 11 categories – Administration, Business, Cultural, Engineering, Finance, Health, ICT, Marketing, Social and Teaching Professionals as well as Business Service Agents. Students, National Service Personnel and Others are occupations which have no clearly defined categories or had minimal representation in the data collected.

From the data, students made up to 30.5% of the patrons of Uber services. Others (13.3%) entails of but not limited to Photographers, Graphic Designers, Masons, Logisticians, Volunteers, Artists, Public Servants, Drivers, Researchers, Shoe Manufacturers and Office Assistants. National Service Personnel made up 11.9% while Teaching Professional (3.4%) made up of lecturers and teachers constituted 3.4% of the total study. The unemployed also made the list representing 1.7%.

Table 4: Occupation of riders on frequency of use of Uber Services

	Uber Patronage Frequency			
Occupation	**Frequently**	**Less frequent**	**Very frequent**	**Total**
Administration Professional	7	1	2	10
Business Professional	3	9	3	15
Business Service Agents	3	2	2	7
Cultural Professional	10	4	0	14
Engineering Professional	7	7	2	16
Finance Professional	14	18	5	37
Health Professional	3	8	4	15
ICT Professional	6	2	0	8
Marketing Professional	4	7	0	11
National Service Personnel	12	26	4	42
Other	21	17	9	47
Social Professional	2	3	1	6
Student	36	64	8	108
Teaching Professional	4	5	3	12
Unemployed	1	4	1	6
TOTAL	**133**	**177**	**44**	**354**

Source: Fieldwork (2018).

The study further sought to establish the relationship between respondents' occupation and how frequently they patronised Uber services. Using confidence level

($\alpha = 0.05$), a chi-square value of 43.013 with a p-value of 0.035 was obtained. The relationship therefore is significant since p-value (0.035) < 0.05. From the table, majority of the respondents patronise Uber frequently with the dominants being Students. However, the definition of frequency by respondents is subjective.

4.1.4 Occupation of Uber drivers

Table 5: Types of employment of Uber drivers

Employment type	Frequency	Percent
Full time	27	64.3
Part time	15	35.7
Total	**42**	**100.0**

Source: Fieldwork (2018).

With Uber being a disruptive business, the study investigated if Uber drivers were fully or partially employed by Uber and also to know what their other or previous jobs were. Table 5 shows that 64.3% of the respondents were Full time Uber drivers and 35.7% were into Uber as a part time trade.

Table 6: Previous and other jobs of Uber drivers

Part Time (Other Jobs)	Full Time (Previous Jobs)
Electronics Installer	Building Construction
Tertiary Student	Taxi driver
Shop Attendant	Sales representative
Tertiary Student	Mechanic - JHS
Surveyor	Insurance Sales Manager
Real Estate	National Service Personnel
Site supervisor	Trader
Estate Developer	Footballer
Student	Laborer
Sales Executive	Excavators Operator
Marketer	Accountant
Sales Manager	Trotro driver

Gas Station Attendant	Shop owner
	Public servant
	Customer Service Agent
	Hotel Worker
	Tipper Truck Driver
	Distributor

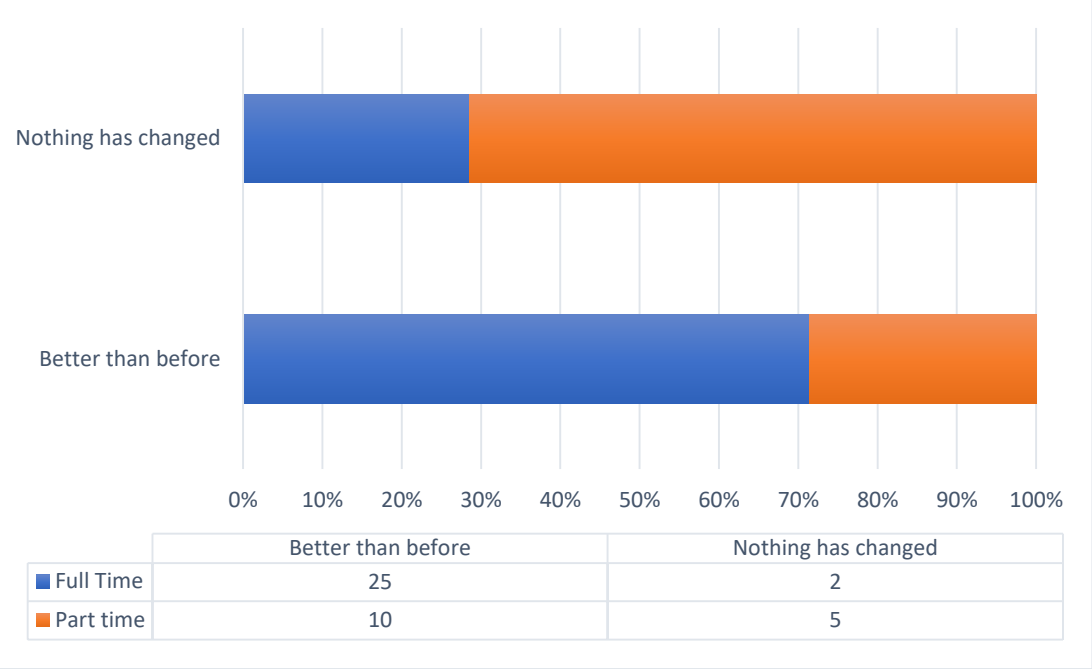
Source: Fieldwork (2018).

The study further sought to enquire what their previous jobs were if they into Uber full time and if part time, what their other jobs are. Table 6 shows that those into Uber full time were formerly Accountants, Contractors, Customer Service Agents, Distributors, Excavator Operators, Hotel workers, Insurance Brokers, Mechanics, Labourers, sales Personnel, National Service Personnel and Traders. Interestingly, the study had 10 respondents who were formerly taxi drivers and a private intra-city bus driver (trotro driver) who were very resourceful to the study because they were in a better position to give the researcher a comparative analysis between driving Uber and a taxi.

The study identified respondents who operated Uber as a part time job who were also into other professions, such as estate developers, marketing executives, electricians, sales personnel, students and surveyors.

4.2 Uber's Livelihood Impact

Figure 1: Livelihood impacts of Uber drivers



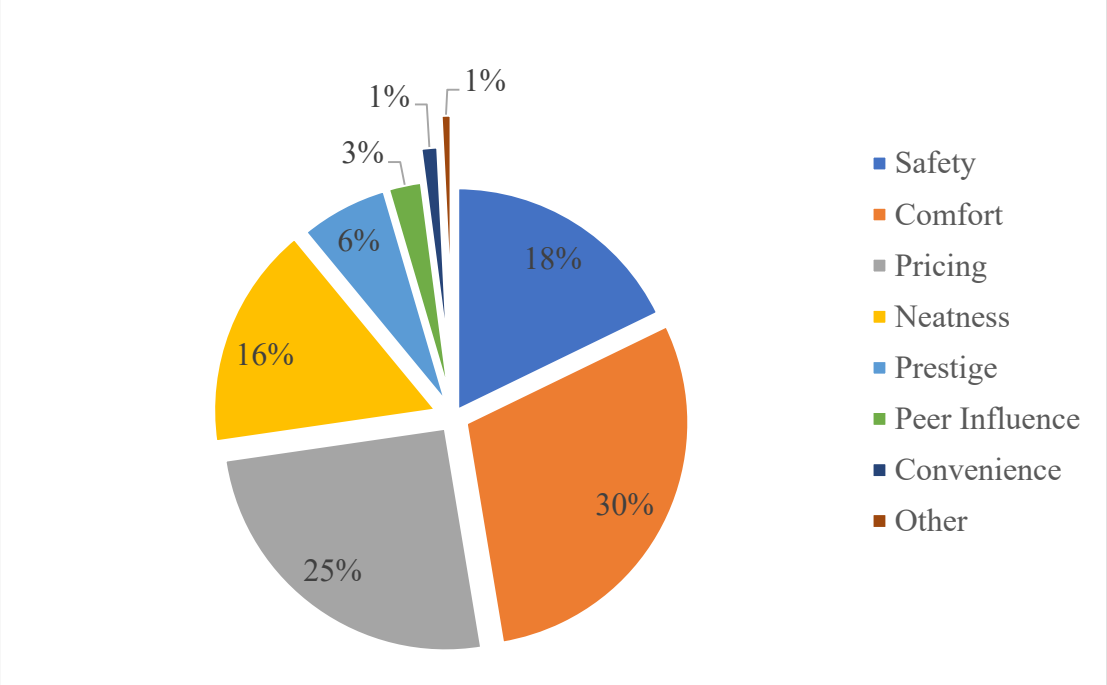
Source: Fieldwork (2018).

In order to assess the impact of Uber on the livelihood of the drivers, respondents were asked to choose amongst three options – Better than before, Nothing has changed and Worse than before. Figure 1 shows that 85.4% of respondents' livelihood have been improved since they started working with Uber while 14.6% claimed nothing had really changed.

Those who responded that their livelihoods had improved attributed it to flexible job schedules, improved income and the opportunity it creates for networking. Some respondents added that there was prestige in being an Uber driver because a lot of people in good paying jobs are also driving Uber. Those who said nothing had changed attributed it to the fact that their jobs do pay better and income from Uber is just residual. Some also indicated they loved to drive and just needed to make profit from it in order not to waste fuel.

4.3 Uber Patronage Factors

Figure 2: Factors for Uber patronage



Source: Fieldwork (2018).

Natyari & Pradana (2016) identified five factors - tariffs, practicality, trust, comfort, and safety, as the influencers of consumers' choice of Uber in the city of Bandung with comfort being the dominant factor. This is no different from what this study gathered as comfort (29.6%) is also the dominant pull factor of Uber patronage in the Accra Metropolis. However, new factors were identified as patronage influencers and they include Prestige (6.4%) and Peer Influence (2.5%). Others (0.8%) constituted Driver Professionalism, Discounts and Unavailability of public transport.

Comfortability being the dominant pull factor is as a result of the Air-condition riders enjoy through the journey upon request. Some Uber drivers go a step further to show movies for their riders and sometimes ask riders to select a song of their choice. The professionalism displayed by drivers is one of the advantages Uber has over taxis. Wallsten (2015) showed that Taxi drivers in New York and Chicago responded to competition by improving quality. This was not the case for any of the respondents –

taxi drivers, who partook in the study. Drivers and riders alike, rate each other on a 5-star rating scale. The higher the ratings, the more beneficial it is for a driver. Drivers know the essence of these ratings and therefore encourage their passengers to give them good ratings. They in return also try to live up to the task. Uber also gives out discounts to its patrons and this ensures they always come back. Hence, a taxi driver's charges may always seem higher even if it is a normal fare.

4.4 Taxi Patronage Factors

Table 7: Taxi patronage factors

Factors	Frequency	Percent
Emergency Situation	229	40.7
Price Hikes in Uber Fares	143	25.4
Road Traffic Congestion	180	32.0
Taxi Availability	8	1.4
Others	2	0.4
Total	**562**	**100.0**

Source: Fieldwork (2018).

The study also sought to enquire what factors influence people's use of taxis over Uber. Emergency situation (40.7%) was the dominant factor. In situations where one urgently had to move from one place to the other, most users sought the services of taxis where they negotiate the fares and quickly join the taxis. This is attributed to the fact that Uber cars upon request may be some minutes away and may delay the rider's schedule.

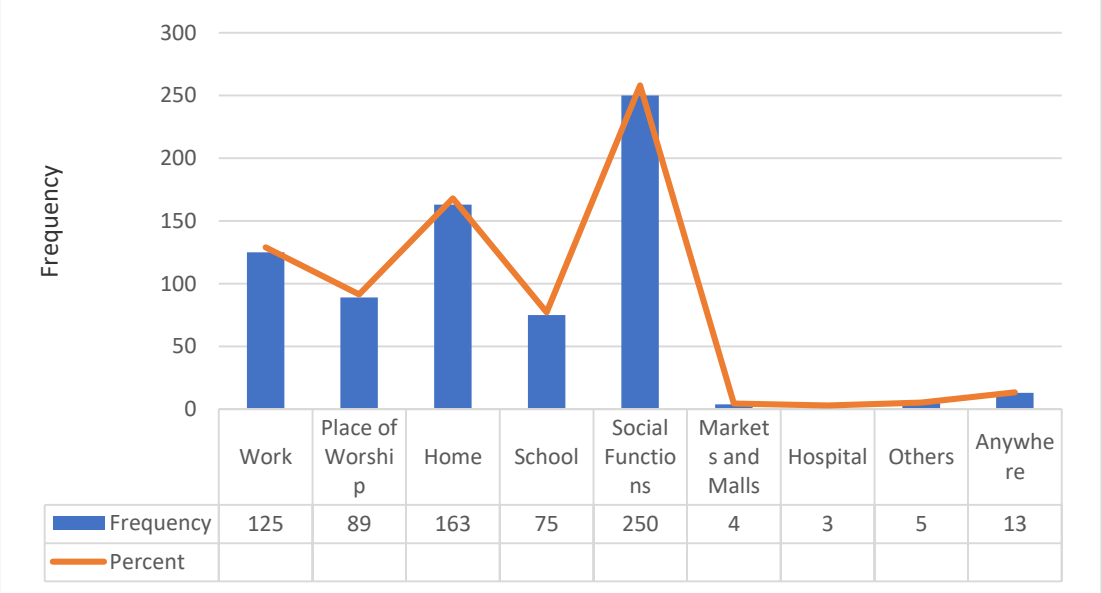
Road traffic congestions (32%) is the next factor followed by price hikes (25.4%) and this is as a result of hikes in Uber fares occurring during road traffic congestion. A hike in the fares of Uber services can be attributed to two factors -

an increase in demand of Uber vehicles and road traffic congestion. When demand is high within the rider's geographical location, fares slightly increase and then stabilize when demand is low. Per the Uber fare formula - GH¢1.70 + GH¢1.30 per kilometre + GH¢0.19 per minute, riders can be faced with price hikes in cases of road traffic congestion because for every minute spent in a road traffic congestion, a rider pays GH¢0.19 more than the actual fare unlike the taxis where negotiated fares remain constant irrespective of distance and hours spent in traffic. This gets most people going in for taxis than Uber especially during festive seasons.

The next factor was availability of taxis (1.4%) where taxis are readily available for the rider, others (0.4%) constituted delay in the arrival of Uber cars and taxis not complaining about the destination. Since one may be in a hurry, a little delay in the stipulated arrival time of requested Uber vehicles makes one opting for a taxi and cancelling the request. Despite paying a cancellation fee of GH¢5 on the next trip, riders still go ahead to cancel. Most Uber drivers upon accepting a rider's request call to find out what their destinations are. For most Uber drivers depending on the nature of the road of the destination or distance to the place, they turn down the request. A taxi driver on the other hand will accept to go to the requested destination and charge equally for it.

4.5 Trip Purpose

Figure 3: Trip purpose

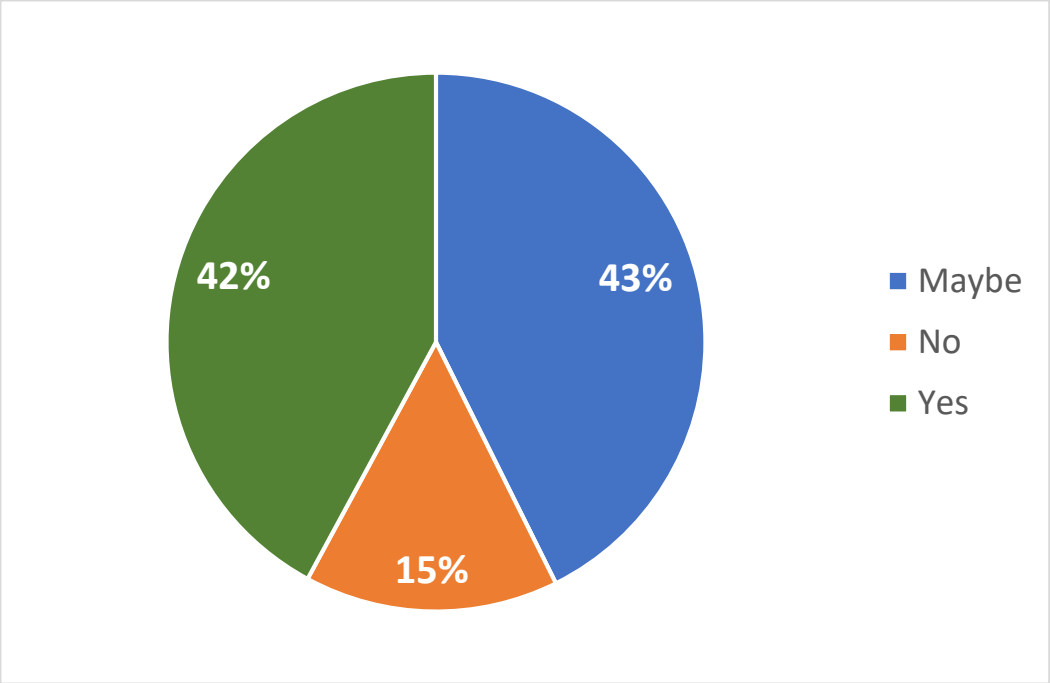


Source: Fieldwork (2018).

The study further investigated the trip purpose of Uber patrons and discovered that about 34.4% made the trips by Uber to attend social functions such as parties, concerts, funerals, weddings and festivals. Trips to school made 10.3% of total trip purpose. 22.4% and 17.2% of Uber's riders made trips by Uber to their homes and workplaces and this can be attributed to the comfort and safety one is highly assured.

4.6 Adoption of Taxis' services

Figure 4: Choice of taxis over Uber rides



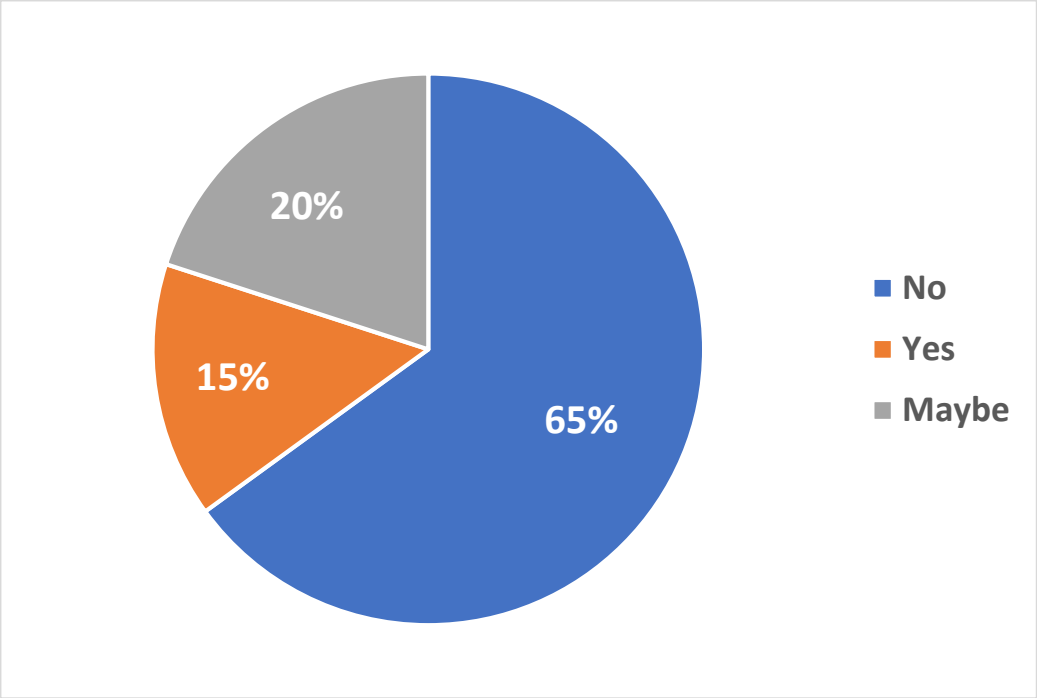
Source: Fieldwork (2018).

This study further enquired from respondents if they will patronise Taxis over Uber especially if taxis adopted the pricing system of Uber - setting a base fare at GH¢1.70 + GH¢1.30 per kilometre and GH¢0.19 per minute, making its minimum fare GH¢5. 42.7% of the respondents chose "maybe" with 42.1% responding in the affirmative. 15.3% of the total sample responded in the negative.

This implies there is a greater probability that taxi drivers will be giving Uber a good run for their money should they adopt this Uber pricing system or an alternative meter pricing system.

4.7 Taxi Drivers Switch to Uber's Pricing System

Figure 5: Probability of adopting Uber's pricing system

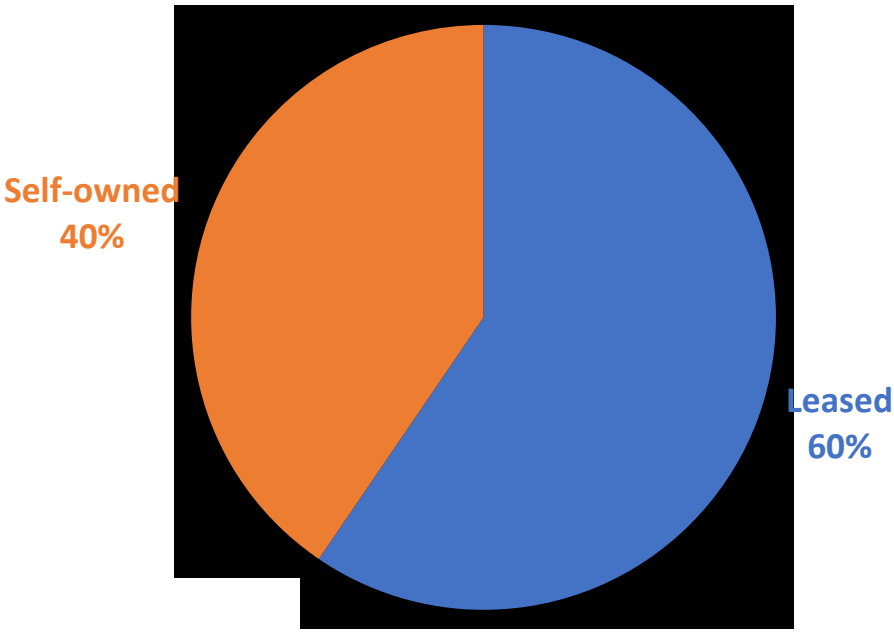


Source: Fieldwork (2018).

Based on the above assertion by Uber riders (Figure 4), the study enquired from taxi drivers if they will be willing to adopt this pricing system. 65% responded in the negative. A common reason cut across their responses upon further discussion which is, the pricing meter will not reflect an increase in fuel prices. Hence, the charges may not make economic sense to them. 4% said 'maybe' with their reason being, only if the regulatory body, Ghana Private Road Transport Union (GPRTU) advices them to do so. 15% responded in the affirmative. To them, once it will bring fairness in price fixing and reduce competition, they accept to adopt the change.

4.8 Vehicle Ownership

Figure 6: Uber vehicle ownership



Source: Field Data, 2018.

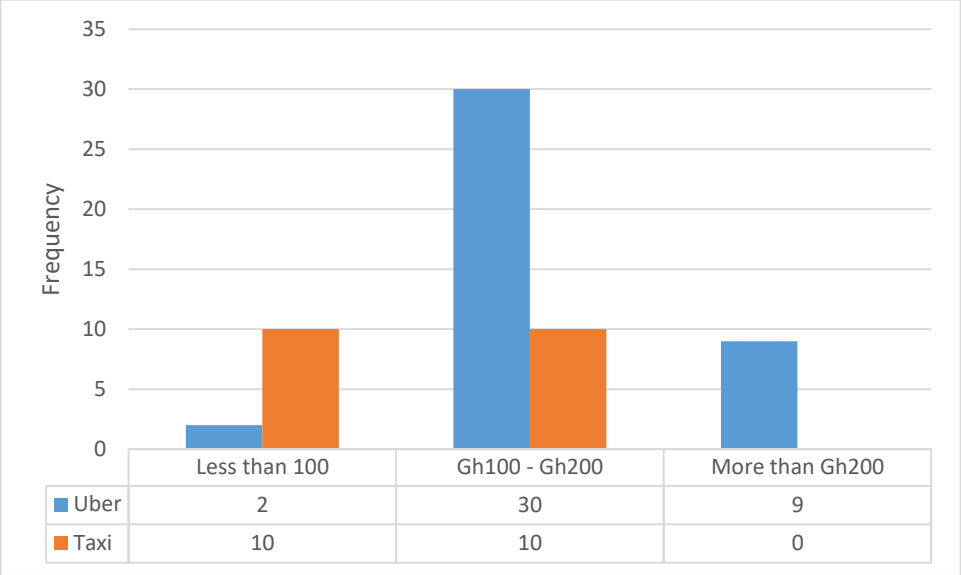
Uber does not give cars to its drivers. Rather, it makes use of people's private cars in its operations. This study enquired the ownership rate of these vehicles used for Uber. It was established that 59.5% of the study's respondents were using leased cars which means the cars they used did not belong to them while 40.5% were the car owners.

With Uber acting as a middle agent between car owners and riders, Uber does not spend money on vehicle maintenance which means money which could have been used for such operation costs are being utilised into something else. Where drivers and car owners feel unfairly treated, they can retract their vehicles and this will have a negative income implication on Uber. In April 2018 when Uber gave discounts of up to 95% off to riders, most Uber drivers the researcher interacted with claimed they parked their vehicles till the issue was resolved and were not questioned by Uber neither were they sanctioned. With this, car owners in a way have some control over

their service provider and it could spell doom for Uber should car owners decide to retreat.

4.9 Income

Figure 7: Income comparison between Uber and Taxis



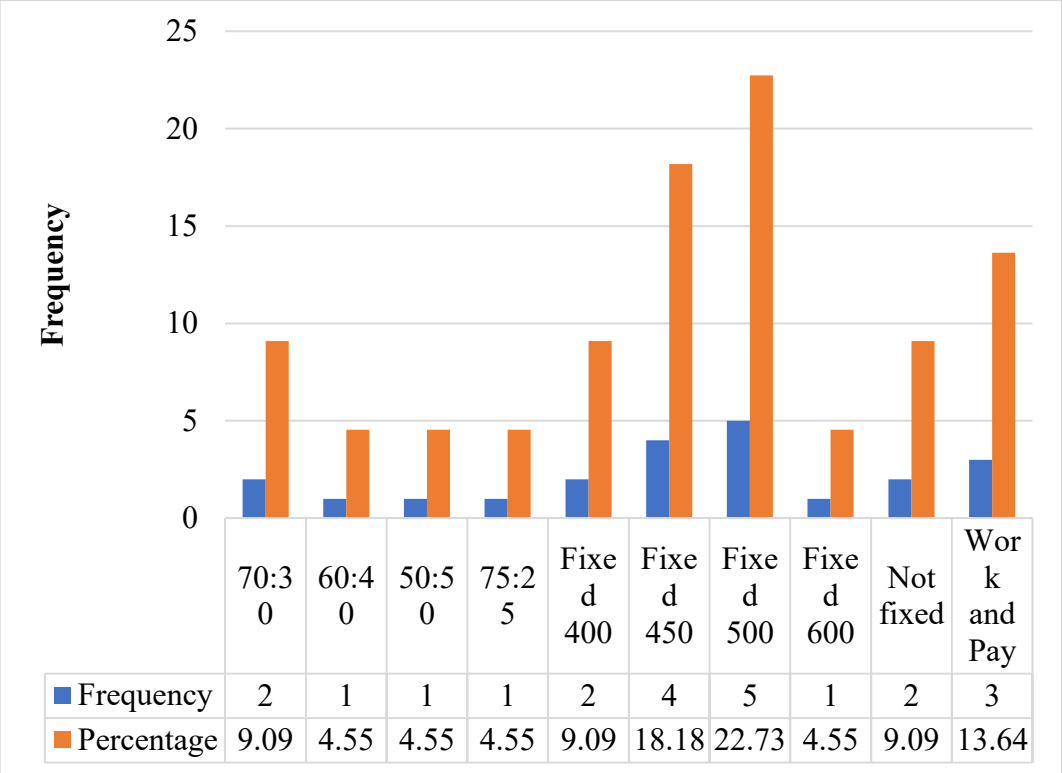
Source: Fieldwork (2018).

Krueger & Cramer (2015:2) stated, "UberX drivers have a substantially higher capacity utilization rate than do taxi drivers in every city except New York, where the utilization rates are very similar. One could argue that it is desirable to compare UberX and taxi drivers during the same period of time, or one could argue that it makes sense to compute the capacity utilization rate for taxis before Uber entered the market to assess the effect of taxi licensing and regulation, because the presence of Uber could have caused the productivity of taxi drivers to change." Capacity utilization is a measure of the extent to which the productive capacity of a business is being used. It can be defined as: The percentage of total capacity that is actually being achieved in a given period (Riley, 2018).

To examine whether the introduction of Uber has had an effect on the operating costs of traditional taxi operators, the study computed the weekly sales levels of taxi drivers, before and after the inception of Uber, to determine capacity utilisation of

Uber and taxis. It analysed the weekly sales levels of Uber and Taxi drivers – stationed and non-stationed drivers and made a comparison between their average weekly sales before and after the inception of Uber based on a sampling of 10 drivers each from stationed and non-stationed drivers. From Figure 7, it can be seen that most Uber drivers earn between Gh¢100 and Gh¢200 daily on the average; and most Taxi drivers do not earn more than Gh¢200 daily from sales. Comparing the two results, it can be deduced Uber drivers earn more than Taxi drivers. This implies Uber has a higher utilization rate than taxi drivers.

Figure 8: Income split between Uber drivers and their car owners

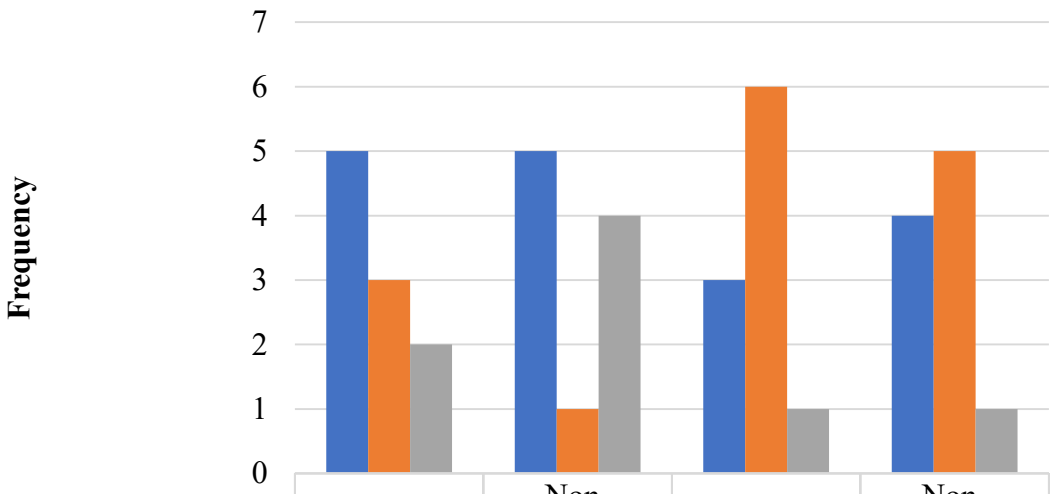


Source: Fieldwork (2018).

Vehicle owners of leased cars also earn significantly from Uber. This came to light from the data gleaned from as respondents on how much they shared with their car owners. It was supposed to be on a ratio scale but some car owners had fixed rates they took per week. From the column chart it can be deduced that car owners earn Gh¢500 a week. Others do not have any fixed amount but a rate so whatever the driver

makes within the week is shared accordingly. Some drivers too are left with the option of giving any amount they want while the rest are on a work and pay scheme.

Figure 9: Average weekly sales comparisons before and after Uber's inception



	Stationed	Non-Stationed	Stationed	Non-Stationed
	Before Uber		After Uber	
Gh¢100 - Gh¢200	5	5	3	4
Less than Gh¢100	3	1	6	5
More than Gh¢200	2	4	1	1

Source: Fieldwork (2018).

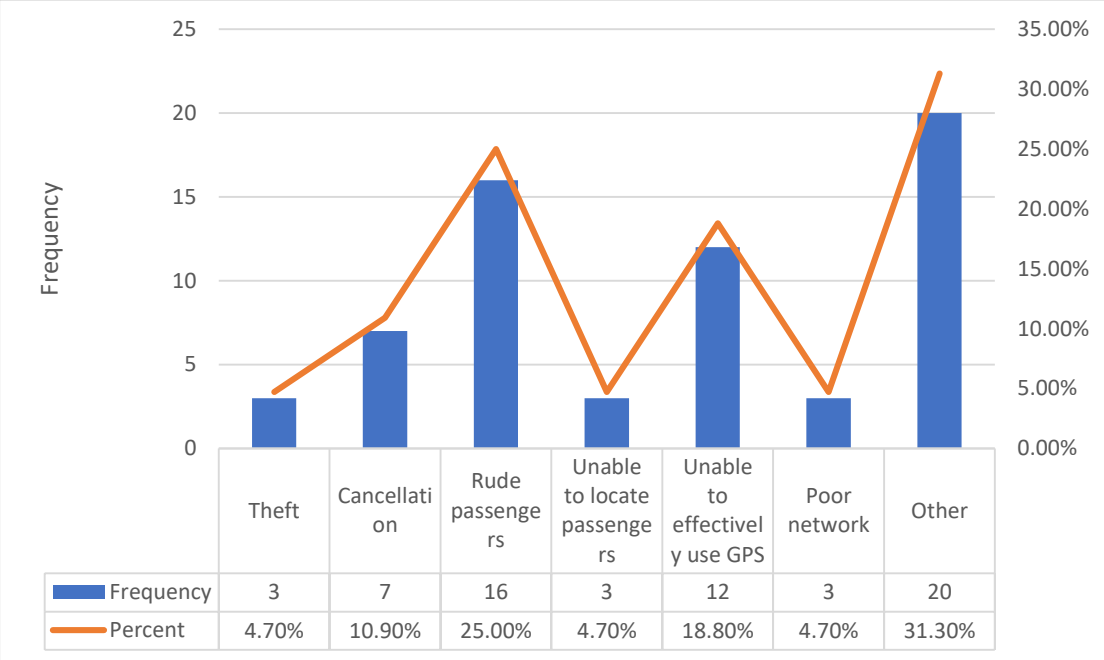
The study went further to explore the average weekly sales of stationed and non-stationed drivers before and after the inception of Uber. It finds in Figure 9 that non-stationed taxi drivers were earning more than stationed taxi drivers before the inception of Uber. It is also observed that before the inception of Uber on the Accra market, most non-stationed and stationed taxi drivers were making weekly average sales of Gh¢100 - Gh¢200 and Gh¢200. The chart shows a significant reduction in how much taxi drivers made averagely weekly. As it shows most of them earned less than Gh¢100 with stationed drivers suffering the big setback.

Interacting with stationed drivers, they attributed it to the fact that people in recent times like to be picked at their doorsteps since the arrival of Uber. Hence, instead of

walking to the station, they either requested for Uber vehicles or stopped taxis in front of their locations.

4.10 Challenges of the Uber driver

Figure 10: Challenges faced by Uber drivers



Source: Fieldwork (2018).

The biggest challenge confronting Uber drivers is the 25% charge Uber deducts per trip and discounts Uber gives its riders and this is represented by others (31.3%). The charge is what Uber takes for providing its service to the driver and this disturbs a lot of Uber drivers as it was major problem which popped up while interviewing them. The researcher however from observation on the field realized some Uber drivers did beat the system. They logged out of the Uber platform and operated as freelancers especially around the city centre. That way, they are able to make sales without being charged 25% by Uber.

At the time of carrying out this research, Uber was giving its riders discounts of up to 95%. This led to a major demonstration by drivers against the company for which Uber partially withdrew the discounts. The reasons for the discount and its later

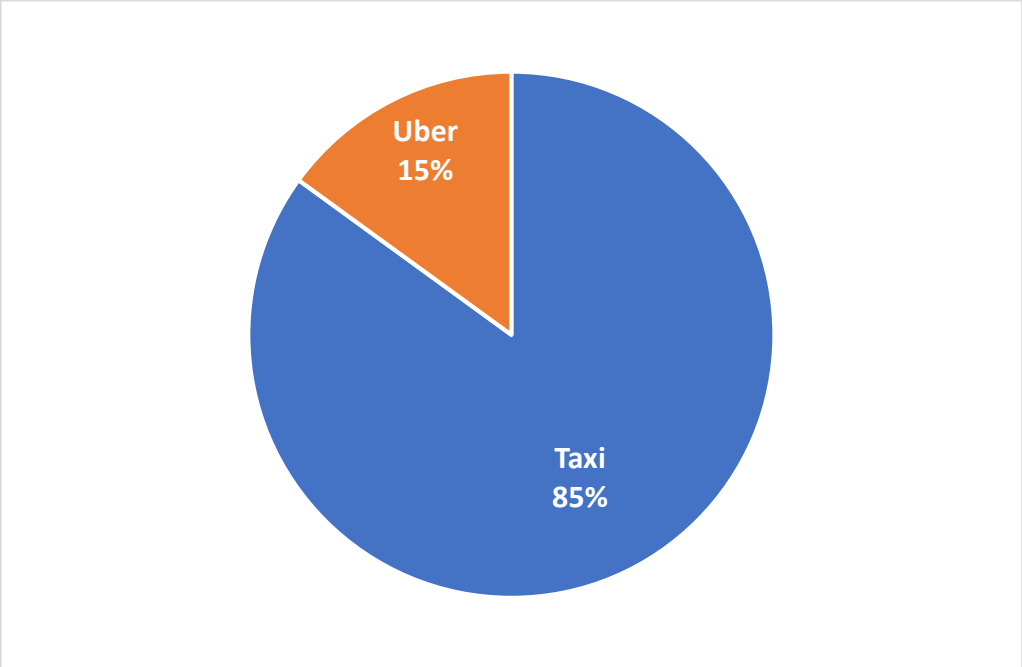
retrieval could not be easily gathered due to the time constraint on the side of the researcher.

The second major problem is their encounter with rude passengers (25%) of which according to some, end up cancelling the request as a way of dealing with the arrogance.

Inability to locate passengers, poor network and theft, were the least of challenges faced by Uber drivers within the study's sample as they all shared a 4.7% proportion of the total challenges.

4.11 Drivers' preference for traditional taxis and Uber services

Figure 11: Taxi drivers' preference for shuttle service operations



Source: Fieldwork (2018).

The study finally asked Taxi drivers to choose between Uber and Taxi. 85% chose taxi while 15% chose Uber. Those who chose Uber said it was a part of their plans under consideration and found it to be more lucrative than the taxi business.

Those who chose Taxi said it was not economical because Uber drivers had to get requests before. They also claimed the 25% charge Uber deducts per trip is a

discouraging factor. Other reasons why they chose taxis over Uber is the fact that taxis are easily identifiable hence believe taxis earn more than Uber drivers. However, a majority attributed it to the fact that they are not technologically savvy and will find it difficult operating the smart devices Uber drivers used.

CHAPTER FIVE

SUMMARY, CONCLUSIONS AND RECOMENDATIONS

5.0 Introduction

This chapter summarises the findings, concludes and makes recommendations based on discussions in the previous chapters.

5.1 Summary of findings

The purpose of this study was to find out the economic impacts of Uber on traditional taxi operators in the Accra Metropolis. Thus the study satisfied the following research objectives:

i. Establish the Uber impact on taxi drivers' market power by measuring the profit margins of traditional taxi operators.

ii. Measure the impact of Uber on traditional taxi operators with different operation models – the unionized taxi operators and the non-unionized taxi operators.

iii. Enquire if traditional taxi operators are willing to adapt the Uber system of pricing.

iv. Investigate the factors which influence people's patronage of the Uber service.

v. Examine whether the introduction of Uber has had an effect on the operating costs of traditional taxi operators.

vi. Determine the categories of people who patronize the service by taking into consideration their age, occupation and sex.

The study was conducted with a sample of 417 respondents – 354 Uber riders, 43 Uber drivers and 20 taxi drivers, who were sampled through e-snowballing and e-accidental sampling techniques. The Statistical Program for Social Science (SPSS)

was used for data capture and analysis and the results were presented as tables, frequencies, and percentage distributions.

The study's findings showed that majority of Uber's patrons and drivers are between the ages of 21 – 30. This can be said translated as Uber having a youthful population in terms of drivers and riders.

The study further shows that majority of the respondents were Full time Uber drivers while others used it as a side job. It also established that should taxis adopt Uber's modus operandi, they are more likely to enjoy an increase in patronage of taxi services.

The study also established that although patrons were willing to patronise taxis should they adopt Uber's pricing system, majority of taxi drivers are not willing to adopt the pricing system due to ignorance on the operation of the pricing system as most were concerned about adjusting fares in case of increments in fuel prices.

The study revealed that Uber drivers compared to taxi earn more income. This implies Uber fetches more income than taxis do.

5.2 Conclusions

Based on the findings of this study, the objectives of this study were categorized into four – background of Uber users, the employment effect, wage effect and the pull factors of Uber and Taxi services. These categories determined if Uber has had an impact on an already existing job – traditional taxis, it assessed if wages have been affected by the influx of Uber on the job market and what factors influence the patronage of Uber to have an impact of the aforementioned objectives.

Employment effect

To measure the employment effect, this study could not get the number of taxi drivers within a defined period. But per the data gathered, it can be deduced that more of the youth are venturing into Uber rather than taxi because of the prestige attached to being an Uber driver. It can also be said that people are moving away from the taxi business and starting a business in Uber as Uber drivers attested to the benefits of driving Uber. The study can however not state with authority if there has been an impact on the labour supply of the taxi industry. It can be concluded that the inception of Uber into the taxi market of Accra has not destructed the already existing job. It has only provided an alternative employment and it is a matter of choice to join the traditional taxi or modern taxi – Uber.

Wage effect

Although taxi drivers have had their jobs untouched, the income has been greatly affected. The study confirms this as decrease in the wage and sales of Taxi drivers is seen in the assessment of the wages or sales of taxi drivers before and after the inception of Uber. Although they agree the introduction of Uber has affected their income, they are not ready to switch. Ignorance on the side of some traditional taxi drivers on the operations of Uber explains their contentment with their current job.

It was also stated in the previous chapter that Uber drivers were cheating the system by going offline and operating as freelance private taxis. Hence, they make some extra income which is not accounted for to Uber. This they attribute to the 25% charge Uber makes per trip. The study also showed that owners of vehicles used in Uber operations are benefitting massively from the service as shown in the previous chapter.

Pull factors of Uber

The study concludes that Uber patrons consider their comfort as the most critical need that informs their use of the service rather than traditional taxis. Peer influence and prestige are among the other factors strongly considered by patrons.

5.3 Recommendations

The researcher makes the following recommendations to inform policy makers to help improve and maintain a healthy competition among Uber drivers and Traditional Taxi operators.

1. Government: The government through the Ministry of Transport should have a body to regulate the fares of both Uber and Taxi operators to ensure transparency. This will reduce the unnecessary price increments especially by taxi drivers when fuel prices go up.
2. Ghana Private Roads Transport Union (GPRTU): GPRTU should adopt a digitised system of price quotation. With the current technological trends and phases occurring in the transport industry, it is only prudent they ensure a way of being trendy.
3. Uber: Uber needs to review its policy on the 25% per trip deduction it makes. This will ensure Uber drivers do not cheat the system. Also, it needs to carry out a massive education for its drivers and members of the public to ensure productivity and fostering of a good relationship between its drivers and customers. In rolling out promotions and discounts, Uber should have a consultative meeting with its drivers to pre-inform them.

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APPENDICES

Appendix 1: UBER RIDERS QUESTIONNAIRE

UNIVERSITY OF CAPE COAST

FACULTY OF SOCIAL SCIENCE

DEPARTMENT OF GEOGRAPHY AND REGIONAL PLANNING

This questionnaire is a part of a larger study which is assessing ***The Economic Impacts of UBER Services on Traditional Taxi Operators in Accra Metropolis.*** *The respondent's participation will be much appreciated to understand and make suggestions on the above stated topic. Any information provided here is confidential and will be used solely for the purpose of this study. Thank you.*

1. Sex a. Male [ ] b. Female [ ]

2. Age
   A. 10 – 20 **[ ]**
   B. 21 – 30 **[ ]**
   C. 31 – 40 **[ ]**
   D. 41 – 50 **[ ]**
   E. 51 – 60 **[ ]**
   F. 61+ **[ ]**

3. Marital Status
   A. Single **[ ]**
   B. Married **[ ]**
   C. Divorced **[ ]**
   D. Widowed **[ ]**
   E. Co-habitating **[ ]**

4. Religion
   A. Christianity **[ ]** B. Islam **[ ]** C. African Traditional **[ ]** D. Other**[ ]** (Specify……)

5. Number of dependents ……

6. Occupation ……

7. How often do you patronise Uber Services?
   A. Less Frequent **[ ]**
   B. Frequently **[ ]**
   C. Very Frequent **[ ]**

8. What times of the day do you patronise Uber Services the **most**? (Tick as many as applicable)
   A. Dawn **[ ]**
   B. Morning **[ ]**

C. Afternoon **[ ]**
D. Evening **[ ]**
E. Mid-night **[ ]**

9. Which day(s) of the week do you patronise Uber Services the **most**?
   A. Weekdays **[ ]**
   B. Weekends **[ ]**

10. To where do you mostly use UBER services? (**Tick as many as possible)**
    A. Work **[ ]**
    B. Church **[ ]**
    C. School **[ ]**
    D. Lectures **[ ]**
    E. Social Events **[ ]**
    F. Others (Please state……)

11. What factors influence your patronage of UBER? **(Tick as many as applicable)**
    A. Safety **[ ]**
    B. Comfortability **[ ]**
    C. Pricing System **[ ]**
    D. Neat Uber Vehicles **[ ]**
    E. Prestige **[ ]**
    F. Peer Influence **[ ]**
    G. Other (Please state…….)

12. Under what circumstances will you choose Traditional Taxis over Uber? **(Tick as many as applicable)**
    A. Emergency Situations **[ ]**
    B. Price Hikes **[ ]**
    C. Heavy Vehicular Traffic **[ ]**
    D. Others (Please state…….)

13. Will you patronise Traditional Taxi frequently if they adopted the UBER Pricing System?
    A. Yes **[ ]** B. No **[ ]** C. Maybe **[ ]**

14. What is your take on the introduction of UBER and its impact on traditional taxis?

……
……
……

Appendix 2: Interview Schedule for Taxi Drivers

UNIVERSITY OF CAPE COAST

FACULTY OF SOCIAL SCIENCE

DEPARTMENT OF GEOGRAPHY AND REGIONAL PLANNING

The Economic Impacts of UBER Services on Traditional Taxi Operators in Accra Metropolis

This questionnaire is a part of a larger study which is assessing ***The Economic Impacts of UBER Services on Traditional Taxi Operators in Accra Metropolis.*** *The respondent's participation will be much appreciated to understand and make suggestions on the above stated topic. Any information provided here is confidential and will be used solely for the purpose of this study. Thank you.*

1. Sex a. Male [ ] b. Female [ ]

2. Age
   A. 18 – 30 [ ]
   B. 31 – 40 [ ]
   C. 41 – 50 [ ]
   D. 50+ [ ]

3. Marital Status
   A. Single [ ]
   B. Married [ ]
   C. Divorced [ ]
   D. Widowed [ ]
   E. Co-habitating [ ]

4. Religion
   A. Christianity [ ] B. Islam [ ] C. African Traditional [ ] D. Other [ ] (Specify……)

5. Number of dependents ……

6. How long have you been driving Taxi?
   A. Less than a year [ ]
   B. 1 – 5 years [ ]
   C. 5 years+ [ ]
7. Have you heard of Uber?
   D. Yes [ ]
   E. No [ ]

8. Are you working with your taxi full time?
   A. Yes **[ ]** B. No [ ]

9. What is the vehicle's ownership?
   F. Self-owned **[ ]**
   G. Employed **[ ]**

   9.1. If employed, how much do you earn monthly? ……

10. How much do you earn on an average daily?
   B. Less than Gh¢100 **[ ]**
   C. Gh¢100 - Gh¢200 **[ ]**
   D. More than Gh¢200 **[ ]**

11. Which day(s) of the week do you provide taxi services the **most**?
   C. Weekdays **[ ]**
   D. Weekends **[ ]**

12. What times of the day are you patronised the **most**? (**Tick as many as applicable**)
   A. Dawn **[ ]**
   B. Morning **[ ]**
   C. Afternoon **[ ]**
   D. Evening **[ ]**
   E. Mid-night **[ ]**

13. What are some of the challenges you face since the inception of UBER?
   ……
   ……
   ……

14. How much were you making averagely in a week before Uber came?
   A. Less than Gh¢100 **[ ]**
   B. Gh¢100 - Gh¢200 **[ ]**
   C. More than Gh¢200 **[ ]**

15. How much do you make now averagely in a week?
   D. Less than Gh¢100 **[ ]**
   E. Gh¢100 - Gh¢200 **[ ]**
   F. More than Gh¢200 **[ ]**

16. Comparing the two services – UBER and Taxi, which one do you think is better?
    A. Taxi [ ]
    B. Uber [ ]

    State your reason for the option chosen in Question 16
    ……
    ……
    ……
    ……
17. Are you ready to adopt Uber's system of pricing to your service?
    A. Yes [ ]
    B. No [ ]

    State your reason(s) for the options chosen in Question 17
    ……
    ……
    ……
    ……

Appendix 3: Interview Schedule for Uber Drivers

UNIVERSITY OF CAPE COAST

FACULTY OF SOCIAL SCIENCE

DEPARTMENT OF GEOGRAPHY AND REGIONAL PLANNING

This questionnaire is a part of a larger study which is assessing ***The Economic Impacts of UBER Services on Traditional Taxi Operators in Accra Metropolis.*** *The respondent's participation will be much appreciated to understand and make suggestions on the above stated topic. Any information provided here is confidential and will be used solely for the purpose of this study. Thank you.*

1. Sex a. Male [ ] b. Female [ ]

2. Age
   A. 18 – 30 [ ]
   B. 31 – 40 [ ]
   C. 41 – 50 [ ]
   D. 50+ [ ]

3. Marital Status
   A. Single [ ]
   B. Married [ ]
   C. Divorced [ ]
   D. Widowed [ ]
   E. Co-habitating [ ]

4. Religion
   A. Christianity [ ] B. Islam [ ] C. African Traditional [ ] D. Other [ ] (Specify……)

5. Number of dependents ……

6. How long have you been driving Uber?
   A. Less than a year [ ]
   B. A year [ ]
   C. More than a year [ ]

7. Are you working with Uber full time or part time?
   A. Full Time [ ]
   B. Part Time [ ]

   7.1 If full time, what was your previous occupation?.....

7.2 If part time, what is your other job? …….

8 What is the vehicle's ownership?
   A. Self-owned **[ ]**
   B. Leased **[ ]**

9 If leased, what is the lease arrangement?
   A. Driver takes all **[ ]**
   B. Driver and Owner share proceeds (state **Driver-Owner Ratio** ……**:**……)
   C. Owner takes all **[ ]**

10 How much do you earn on an average daily?
   A. Less than Gh¢100 **[ ]**
   B. Gh¢100 - Gh¢200 **[ ]**
   C. More than Gh¢200 **[ ]**

11 Which day(s) of the week do you provide Uber Services the **most**?
   A. Weekdays **[ ]**
   B. Weekends **[ ]**

12 What times of the day are you patronised the **most**? (**Tick as many as applicable**)
   A. Dawn **[ ]**
   B. Morning **[ ]**
   C. Afternoon **[ ]**
   D. Evening **[ ]**
   E. Mid-night **[ ]**

13 What are the challenges you face as a Uber Driver? **(Tick as many as applicable)**
   A. Theft **[ ]**
   B. Confrontation from Taxi drivers **[ ]**
   C. Cancellation of trips **[ ]**
   D. Rude Passengers [ ]
   E. Unable to locate passengers **[ ]**
   F. Poor Network Reception **[ ]**
   G. Other (Please state…….)

14 Under what circumstances do you make poor sales? **(Tick as many as applicable)**
   A. Holidays **[ ]**
   B. Heavy Vehicular Traffic **[ ]**
   C. Others (Please state…….)

15 Under what circumstances do you make good sales? **(Tick as many as applicable)**
   A. Holidays **[ ]**

B. Heavy Vehicular Traffic **[ ]**
C. Others (Please state…….)

16 How has Uber impacted your livelihood?

A. Better than before **[ ]** B. Nothing has changed **[ ]** C. Worse than before **[ ]**

State Reason for option chosen

……

……

……

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